

Volatility models for REMX: GARCH, GARCH-X, GARCHAND, GARCHND and EGARCH-X

Summary of all models estimated in the notebooks **03-GP-models-gaussian**, **03-GP-models-student** and **04-extension-models**. Each section contains the variance expression, a brief description of what the model is meant to demonstrate, and the main estimation results (Gaussian first, Student-t second). The *Sig.* column summarises significance: *** = 1%, ** = 5%, * = 10%. A final part covers the seven robustness checks from notebooks **06-robustness-checks** and **07-extra-robustness-checks**.

1. GARCH(1,1) — Baseline

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2$$

What we want to demonstrate: Canonical Bollerslev (1986) specification. It serves as the benchmark against which every sentiment-augmented model is compared. The goal is to confirm persistence ($\alpha + \beta$ close to 1) and that the basic parameters are significant; all later extensions are evaluated relative to this fit.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0857	0.0301	2.8497	0.0044	***
alpha[1]	0.0893	0.0161	5.5536	0.0000	***
beta[1]	0.8975	0.0182	49.2043	0.0000	***

Log-likelihood : -5982.3069

AIC : 11970.6138

BIC : 11988.3882

alpha[1] + beta[1] = 0.9867 (stationary (<1))

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0878	0.0299	2.9308	0.0034	***
alpha[1]	0.0841	0.0150	5.5880	0.0000	***
beta[1]	0.9010	0.0178	50.6845	0.0000	***
nu	11.1674	2.3239	4.8054	0.0000	***

Log-likelihood : -5956.2330

AIC : 11920.4660

BIC : 11944.1652

alpha[1] + beta[1] = 0.9850 (stationary (<1))

nu = 11.1674 (smaller -> heavier tails; Gaussian as nu -> infinity)

2. GARCH-X (tone) — G&P Eq. (4)

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma \cdot \text{tone}_{t-1}^2$$

What we want to demonstrate: Symmetric extension that injects the daily average news tone as an exogenous regressor in the variance equation. We want to test whether news intensity (regardless of sign) generates additional volatility in REMX, i.e. whether γ is statistically significant.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0907	0.0305	2.9703	0.0030	***
alpha	0.0898	0.0162	5.5564	0.0000	***
beta	0.8967	0.0184	48.8103	0.0000	***
gamma	-0.0047	0.0151	-0.3121	0.7549	

Log-likelihood : -5982.4884

AIC : 11972.9767

BIC : 11996.6759

alpha + beta : 0.9866 (stationary (<1))

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0953	0.0313	3.0451	0.0023	***
alpha	0.0844	0.0151	5.6021	0.0000	***
beta	0.9005	0.0178	50.6344	0.0000	***
gamma	-0.0078	0.0160	-0.4866	0.6266	
nu	11.0876	2.2973	4.8265	0.0000	***

Log-likelihood : -5956.1902

AIC : 11922.3804

BIC : 11952.0044

alpha + beta : 0.9849 (stationary (<1))

Estimated degrees of freedom: nu_hat = 11.088

Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).

3. GARCH-X (log article growth) — G&P Eq. (4)

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma \cdot \log_art_growth_{t-1}^2$$

What we want to demonstrate: Same structure as the previous GARCH-X, but using the log-growth in article volume ($\log(1+N_t) - \log(1+N_{t-1})$) as a proxy for news intensity. The log transform replaces the raw growth rate to remove a single-day outlier (art_growth = 61 on 2015-10-23) that would otherwise dominate the squared term. We want to see whether sustained changes in media coverage (a significant γ) anticipate next-day volatility.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0780	0.0370	2.1088	0.0350	**
alpha	0.0908	0.0160	5.6643	0.0000	***
beta	0.8961	0.0181	49.5127	0.0000	***
gamma	0.0170	0.0419	0.4068	0.6842	

Log-likelihood : -5982.4084

AIC : 11972.8167

BIC : 11996.5159

alpha + beta : 0.9869 (stationary (<1))

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0894	0.0358	2.4988	0.0125	**
alpha	0.0852	0.0149	5.7116	0.0000	***
beta	0.8996	0.0176	51.0750	0.0000	***
gamma	-0.0003	0.0330	-0.0077	0.9939	
nu	11.1163	2.3046	4.8235	0.0000	***

Log-likelihood : -5956.3184

AIC : 11922.6368

BIC : 11952.2608

alpha + beta : 0.9848 (stationary (<1))

Estimated degrees of freedom: nu_hat = 11.116

Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).

4. GARCHAND (tone) — G&P Eq. (5)

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma (1 + d_1 \cdot \theta) \cdot \text{tone}_{t-1}^2, d_1 = 1\{\text{tone}_{t-1} < 0\}$$

What we want to demonstrate: Asymmetric version of GARCH-X: when the previous day's tone is negative, its impact on variance is amplified by the factor $(1 + \theta)$. We want to demonstrate that negative news drives a stronger volatility response than positive news ($\theta > 0$ and significant).

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0899	0.0319	2.8172	0.0048	***
alpha	0.0899	0.0160	5.6072	0.0000	***
beta	0.8967	0.0181	49.4677	0.0000	***
gamma	-0.0030	0.0162	-0.1857	0.8527	
theta	0.4971	7.5039	0.0663	0.9472	

Log-likelihood : -5982.4997

AIC : 11974.9994

BIC : 12004.6234

alpha + beta : 0.9866 (stationary (<1))

Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.390

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0947	0.0307	3.0788	0.0021	***
alpha	0.0844	0.0151	5.6078	0.0000	***
beta	0.9005	0.0178	50.6732	0.0000	***
gamma	-0.0057	0.0165	-0.3462	0.7292	
theta	0.4959	4.0394	0.1228	0.9023	
nu	11.0871	2.2980	4.8246	0.0000	***

Log-likelihood : -5956.1746

AIC : 11924.3493

BIC : 11959.8980

alpha + beta : 0.9849 (stationary (<1))

Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.390

Estimated degrees of freedom: nu_hat = 11.087

5. GARCHAND (log article growth) — G&P Eq. (6)

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma \cdot d_2 \cdot \log_art_growth_{t-1}^2, d_2 = 1\{\log_art_growth_{t-1} > 0\}$$

What we want to demonstrate: The article-volume effect is switched on only on days in which the news flow GROWS relative to the previous day ($\log_art_growth > 0$). We want to show that novelty (rising coverage) is what moves volatility, rather than low news volumes.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0805	0.0369	2.1798	0.0293	**
alpha	0.0903	0.0159	5.6691	0.0000	***
beta	0.8964	0.0181	49.4928	0.0000	***
gamma	0.0272	0.0936	0.2904	0.7715	

Log-likelihood : -5982.4728

AIC : 11972.9457

BIC : 11996.6449

alpha + beta : 0.9866 (stationary (<1))

Fraction of days with $\log_art_growth_{\{t-1\}} > 0$ ($d_2 = 1$): 0.473

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0938	0.0383	2.4516	0.0142	**
alpha	0.0875	0.0156	5.6277	0.0000	***
beta	0.8994	0.0181	49.6810	0.0000	***
gamma	0.0002	0.0850	0.0022	0.9983	
nu	8.0077	0.7625	10.5017	0.0000	***

Log-likelihood : -5958.4131

AIC : 11926.8262

BIC : 11956.4501

alpha + beta : 0.9869 (stationary (<1))

Fraction of days with $\log_art_growth_{\{t-1\}} > 0$ ($d_2 = 1$): 0.473

Estimated degrees of freedom: $\nu_{\text{hat}} = 8.008$

6. GARCHND (tone & log article growth) — G&P Eq. (7)

Conditional variance:

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma \cdot d_3 \cdot x_{t-1}^2, d_3 = 1\{\sigma_{t-1}^2 \geq \kappa\}, x \in \{\text{tone}, \text{log_art_growth}\}$$

What we want to demonstrate: The news impact activates only when the previous day's variance exceeds a threshold κ , calibrated from an annualised volatility level (30% and 50%). We want to test whether the sentiment effect is non-linear in the volatility regime: a significant γ in the high-volatility regime would imply that news amplifies volatility only when the market is already stressed.

Gaussian

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0929	0.0332	2.8029	0.0051	***
alpha	0.0908	0.0173	5.2433	0.0000	***
beta	0.8920	0.0208	42.9489	0.0000	***
gamma	0.0213	0.0239	0.8914	0.3727	

Log-likelihood : -5982.0113

AIC : 11972.0227

BIC : 11995.7219

alpha + beta : 0.9828 (stationary (<1))

d3 active in-sample ($\sigma_{t-1}^2 \geq \kappa$): 0.616

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0412	0.0243	1.6921	0.0906	*
alpha	0.0845	0.0077	11.0197	0.0000	***
beta	0.9162	0.0086	106.8527	0.0000	***
gamma	-1.2195	0.1779	-6.8566	0.0000	***

Log-likelihood : -5973.4457

AIC : 11954.8915

BIC : 11978.5907

alpha + beta : 1.0008 (non-stationary (>=1))

d3 active in-sample ($\sigma_{t-1}^2 \geq \kappa$): 0.059

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1093	0.0329	3.3242	0.0009	***
alpha	0.0872	0.0117	7.4328	0.0000	***
beta	0.8818	0.0218	40.4393	0.0000	***
gamma	0.2302	0.1017	2.2641	0.0236	**

Log-likelihood : -5979.0497

AIC : 11966.0994

BIC : 11989.7986

alpha + beta : 0.9690 (stationary (<1))

d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.634

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0565	0.0317	1.7793	0.0752	*
alpha	0.0927	0.0093	9.9933	0.0000	***
beta	0.9051	0.0166	54.4964	0.0000	***
gamma	-1.2088	0.1894	-6.3837	0.0000	***

Log-likelihood : -5979.0803

AIC : 11966.1606

BIC : 11989.8598

alpha + beta : 0.9977 (stationary (<1))

d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.075

Student-t

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0919	0.0304	3.0267	0.0025	***
alpha	0.0852	0.0146	5.8442	0.0000	***
beta	0.8977	0.0181	49.6099	0.0000	***
gamma	0.0114	0.0215	0.5333	0.5938	
nu	11.1736	2.3005	4.8571	0.0000	***

Log-likelihood : -5956.1755

AIC : 11922.3509

BIC : 11951.9749

alpha + beta : 0.9829 (stationary (<1))

d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.614

Estimated degrees of freedom: $\nu_{\text{hat}} = 11.174$

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0421	0.0191	2.2089	0.0272	**
alpha	0.0614	0.0049	12.5824	0.0000	***
beta	0.9331	0.0097	96.3702	0.0000	***
gamma	-0.8868	0.0679	-13.0521	0.0000	***
nu	11.9034	2.4404	4.8777	0.0000	***

Log-likelihood : -5951.4249

AIC : 11912.8498

BIC : 11942.4738

alpha + beta : 0.9944 (stationary (<1))
d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.046
Estimated degrees of freedom: $\nu_{\text{hat}} = 11.903$

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0999	0.0314	3.1761	0.0015	***
alpha	0.0845	0.0140	6.0252	0.0000	***
beta	0.8940	0.0170	52.7059	0.0000	***
gamma	0.0958	0.0617	1.5524	0.1206	
nu	11.1160	2.2781	4.8796	0.0000	***

Log-likelihood : -5955.5857
AIC : 11921.1715
BIC : 11950.7954
alpha + beta : 0.9785 (stationary (<1))
d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.633
Estimated degrees of freedom: $\nu_{\text{hat}} = 11.116$

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0605	0.0346	1.7519	0.0798	*
alpha	0.0794	0.0113	7.0419	0.0000	***
beta	0.9135	0.0191	47.7192	0.0000	***
gamma	-1.1900	0.2009	-5.9233	0.0000	***
nu	11.3988	2.3566	4.8371	0.0000	***

Log-likelihood : -5953.9383
AIC : 11917.8766
BIC : 11947.5006
alpha + beta : 0.9928 (stationary (<1))
d3 active in-sample ($\sigma^2_{t-1} \geq \kappa$): 0.060
Estimated degrees of freedom: $\nu_{\text{hat}} = 11.399$

7. EGARCH(1,1)-X — Extension model

Conditional variance:

$$\ln(\sigma_t^2) = \omega + \alpha [|z_{t-1}| - E|z_{t-1}|] + \xi \cdot z_{t-1} + \beta \cdot \ln(\sigma_{t-1}^2) + \gamma_1 \cdot \text{tone}_{t-1} + \gamma_2 \cdot \text{log_art_growth}_{t-1}$$

What we want to demonstrate: Log-variance specification (Nelson, 1991) with two exogenous regressors. It (i) guarantees positivity of σ^2 without parameter restrictions, (ii) captures asymmetry through the leverage term ξ (negative shocks weigh more than positive ones), and (iii) allows tone and log_art_growth to enter linearly (not as squares). We want to show that EGARCH-X captures asymmetry more cleanly than GARCHAND, and to test whether γ_1 and γ_2 are significant in log-variance rather than in squared variance.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0302	0.0106	2.8400	0.0045	***
alpha	0.1723	0.0287	6.0117	0.0000	***
xi	-0.0219	0.0129	-1.7070	0.0878	*
beta	0.9816	0.0066	147.7053	0.0000	***
gamma_tone	0.0057	0.0067	0.8509	0.3948	
gamma_log_artg	-0.0412	0.0599	-0.6877	0.4916	

Log-likelihood : -5978.7694

AIC : 11969.5389

BIC : 12005.0876

beta = 0.9816 (stationary (|beta|<1))

Half-life of a log-variance shock: 37.3 trading days

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0242	0.0088	2.7351	0.0062	***
alpha	0.1596	0.0244	6.5356	0.0000	***
xi	-0.0150	0.0104	-1.4504	0.1469	
beta	0.9826	0.0058	168.2762	0.0000	***
gamma_tone	0.0084	0.0054	1.5508	0.1210	
gamma_log_artg	-0.0600	0.0584	-1.0282	0.3039	
nu	10.9620	2.1873	5.0117	0.0000	***

Log-likelihood : -5952.1628

AIC : 11918.3257

BIC : 11959.7992

beta = 0.9826 (stationary (|beta|<1))

Half-life of a log-variance shock: 39.5 trading days

nu = 10.9620 (Student-t dof; lower -> heavier tails. Note: the t-ratio for nu tests H0: nu = 0, which is not the meaningful null. To test Gaussian vs Student-t use an LR test against the Gaussian EGARCH-X.)

Part II — Robustness Checks (Notebooks 06 & 07)

Seven diagnostic checks. Checks 1–5 are from notebook 06; Checks 6–7 from notebook 07. Check 1 was extended to a rolling-window scheme (813 origins, daily refit on 1,260 days, both RV proxies, DM and Clark–West tests) and applied to **all nine augmented candidates plus the GARCH(1,1) baseline**, not only the GARCHND $\kappa=50\%$ pair. The headline result is sharper than the earlier static-OOS reading: only the $\kappa=50\%$ GARCHND specifications add OOS value — the `log_artg` version dominates at $h=1$ and $h=5$ under both proxies (DM and CW $p < 0.001$), the tone version at $h=1$ only, and both fade or invert at $h=22$. The other augmented models (simple GARCH-X, GARCHAND, $\kappa=30\%$ GARCHND) lose to the baseline at every horizon, and EGARCH-X is roughly tied with the baseline. Checks 2–4 and 6 then characterise the in-sample γ in the surviving $\kappa=50\%$ pair: it is partly redundant with a regime intercept (Check 2), moderately sensitive to the sigmoid sharpness (Check 3, $CV \approx 0.3$ across smooth K), driven by ~17–21 high-vol episodes concentrated in 2020-Q1, 2021-Q1 and 2025-Q4 (Check 4), but robust to HAC and moving-block-bootstrap corrections (Check 6). A model-free lead-lag cross-correlation (Check 5) finds at most a marginal news→volatility lead for signed tone (prewhitened $\rho = +0.043$ at $k = -6$) and none for news intensity or article growth. Check 7 documents that the GDELT sentiment series is built from a permissive filter dominated by the ‘cerium’ chemistry token, which puts an obvious noise ceiling on the magnitude of any sentiment effect — consistent with the conclusion that the OOS predictive content sits with the volatility-regime switch, not with raw news intensity.

Check 1 — Rolling-window out-of-sample forecasting ($h = 1, 5, 22$ days)

Out-of-sample window: **2023-01-03 to 2026-03-31** (813 forecast origins). Parameters are re-estimated **daily** on a **1,260-day rolling sample** for every candidate model in the report (GARCH(1,1) baseline plus 9 augmented specs). The forecast at origin τ is the average of the iterated conditional-variance path $\sigma^2_{\tau+1|\tau}, \dots, \sigma^2_{\tau+h|\tau}$; the target is the corresponding h -day mean of the realised-variance proxy. Two proxies are reported in parallel: **Parkinson** (RV_park) and **Rogers–Satchell** (RV_rs), both in $\%^2/\text{day}$. Losses: HRMSE, QLIKE (Patton 2011, robust to RV-noise), RMSE, MAE. Tests against the GARCH(1,1) Student-t baseline use Newey–West LRV at lag $h-1$: **Diebold–Mariano** (two-sided on squared-loss differential; DM $> 0 \Rightarrow$ candidate has lower MSE) and **Clark–West** (one-sided, the appropriate test for nested models since the baseline is $\gamma = 0$; CW $> 0 \Rightarrow$ candidate adds genuine predictive content). Rows sorted by model $\rightarrow$ proxy $\rightarrow h$.

Loss functions (averaged over the 813 origins)

Model	Proxy	h	HRMSE	QLIKE	RMSE	MAE	N
GARCH(1,1)	Parkinson	1	5.4023	2.0760	4.8866	3.9585	813
GARCH(1,1)	Parkinson	5	2.9753	2.0933	4.4833	3.7785	809
GARCH(1,1)	Parkinson	22	2.6582	2.1211	4.1826	3.6778	792
GARCH(1,1)	Rogers-Satchell	1	12.5808	2.0617	4.8178	3.9280	813
GARCH(1,1)	Rogers-Satchell	5	2.9436	2.0791	4.3905	3.7677	809
GARCH(1,1)	Rogers-Satchell	22	2.6068	2.1039	4.1738	3.6484	792
GARCH-X tone	Parkinson	1	5.4271	2.0793	4.9458	4.0022	813
GARCH-X tone	Parkinson	5	2.9987	2.0971	4.5588	3.8321	809
GARCH-X tone	Parkinson	22	2.6664	2.1250	4.2746	3.7617	792
GARCH-X tone	Rogers-Satchell	1	12.6381	2.0646	4.8757	3.9737	813

GARCH-X tone	Rogers-Satchell	5	2.9624	2.0825	4.4649	3.8186	809
GARCH-X tone	Rogers-Satchell	22	2.6122	2.1073	4.2596	3.7338	792
GARCH-X log_artg	Parkinson	1	5.4685	2.0832	4.9566	4.0203	813
GARCH-X log_artg	Parkinson	5	3.0431	2.1041	4.5634	3.8583	809
GARCH-X log_artg	Parkinson	22	2.8084	2.1412	4.3044	3.8024	792
GARCH-X log_artg	Rogers-Satchell	1	12.8300	2.0684	4.8846	3.9919	813
GARCH-X log_artg	Rogers-Satchell	5	3.0172	2.0895	4.4682	3.8420	809
GARCH-X log_artg	Rogers-Satchell	22	2.7595	2.1235	4.2922	3.7602	792
GARCHAND tone	Parkinson	1	5.6238	2.1006	5.0643	4.1395	813
GARCHAND tone	Parkinson	5	3.0913	2.1121	4.6541	3.9394	809
GARCHAND tone	Parkinson	22	2.6320	2.1167	4.2332	3.6960	792
GARCHAND tone	Rogers-Satchell	1	14.5021	2.0868	4.9992	4.1233	813
GARCHAND tone	Rogers-Satchell	5	3.0861	2.0983	4.5657	3.9263	809
GARCHAND tone	Rogers-Satchell	22	2.5894	2.0994	4.2198	3.6645	792
GARCHAND log_artg	Parkinson	1	5.4361	2.0799	4.9393	3.9990	813
GARCHAND log_artg	Parkinson	5	3.0202	2.0993	4.5381	3.8200	809
GARCHAND log_artg	Parkinson	22	2.7417	2.1279	4.2219	3.7027	792
GARCHAND log_artg	Rogers-Satchell	1	12.6889	2.0653	4.8685	3.9704	813
GARCHAND log_artg	Rogers-Satchell	5	2.9907	2.0848	4.4437	3.8048	809
GARCHAND log_artg	Rogers-Satchell	22	2.6878	2.1103	4.2095	3.6630	792
GARCHND $\kappa=30\%$ tone	Parkinson	1	5.4311	2.0787	4.9183	3.9863	813
GARCHND $\kappa=30\%$ tone	Parkinson	5	2.9983	2.0970	4.5302	3.8153	809
GARCHND $\kappa=30\%$ tone	Parkinson	22	2.6775	2.1271	4.2534	3.7571	792
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	1	12.6402	2.0639	4.8479	3.9530	813
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	5	2.9653	2.0825	4.4361	3.8023	809
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	22	2.6262	2.1093	4.2392	3.7267	792
GARCHND $\kappa=30\%$ log_artg	Parkinson	1	5.4728	2.0808	4.9281	3.9974	813
GARCHND $\kappa=30\%$ log_artg	Parkinson	5	3.0606	2.1033	4.5380	3.8333	809
GARCHND $\kappa=30\%$ log_artg	Parkinson	22	2.8800	2.1401	4.3128	3.7856	792
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	1	12.7754	2.0670	4.8603	3.9673	813
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	5	3.0416	2.0901	4.4502	3.8278	809
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	22	2.8225	2.1244	4.3106	3.7493	792
GARCHND $\kappa=50\%$ tone	Parkinson	1	5.0987	2.0599	4.6817	3.8280	813
GARCHND $\kappa=50\%$ tone	Parkinson	5	2.8735	2.0820	4.3645	3.7098	809
GARCHND $\kappa=50\%$ tone	Parkinson	22	2.7304	2.1394	4.5071	3.9632	792
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	1	12.1220	2.0447	4.6146	3.7828	813
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	5	2.8209	2.0673	4.2739	3.7039	809
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	22	2.6711	2.1221	4.4999	3.9479	792
GARCHND $\kappa=50\%$ log_artg	Parkinson	1	5.2228	2.0639	4.5829	3.7801	813

GARCHND $\kappa=50\%$ log_artg	Parkinson	5	2.8647	2.0817	4.1814	3.6072	809
GARCHND $\kappa=50\%$ log_artg	Parkinson	22	2.5875	2.1172	4.0473	3.6191	792
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	1	12.4078	2.0485	4.4993	3.7318	813
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	5	2.8427	2.0666	4.0714	3.5971	809
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	22	2.5565	2.0995	4.0380	3.5954	792
EGARCH-X	Parkinson	1	5.6122	2.1150	4.9407	4.1347	813
EGARCH-X	Parkinson	5	3.0832	2.1221	4.4406	3.8878	809
EGARCH-X	Parkinson	22	2.5222	2.1166	3.9003	3.5732	792
EGARCH-X	Rogers-Satchell	1	15.0765	2.0988	4.8522	4.0859	813
EGARCH-X	Rogers-Satchell	5	3.1362	2.1060	4.3254	3.8659	809
EGARCH-X	Rogers-Satchell	22	2.5188	2.0984	3.8857	3.5367	792

Diebold–Mariano and Clark–West vs GARCH(1,1) baseline

Candidate	Proxy	h	DM	DM p	CW	CW p
GARCH-X tone	Parkinson	1	−7.883	0.0000	−7.768	1.0000
GARCH-X tone	Parkinson	5	−4.738	0.0000	−4.690	1.0000
GARCH-X tone	Parkinson	22	−2.061	0.0393	−2.022	0.9784
GARCH-X tone	Rogers-Satchell	1	−7.647	0.0000	−7.510	1.0000
GARCH-X tone	Rogers-Satchell	5	−4.569	0.0000	−4.503	1.0000
GARCH-X tone	Rogers-Satchell	22	−2.048	0.0405	−1.996	0.9770
GARCH-X log_artg	Parkinson	1	−6.321	0.0000	−5.960	1.0000
GARCH-X log_artg	Parkinson	5	−3.891	0.0001	−3.664	0.9999
GARCH-X log_artg	Parkinson	22	−2.203	0.0276	−1.929	0.9731
GARCH-X log_artg	Rogers-Satchell	1	−5.862	0.0000	−5.479	1.0000
GARCH-X log_artg	Rogers-Satchell	5	−3.689	0.0002	−3.450	0.9997
GARCH-X log_artg	Rogers-Satchell	22	−2.135	0.0328	−1.856	0.9683
GARCHAND tone	Parkinson	1	−12.523	0.0000	−11.809	1.0000
GARCHAND tone	Parkinson	5	−5.973	0.0000	−5.555	1.0000
GARCHAND tone	Parkinson	22	−0.823	0.4103	−0.397	0.6542
GARCHAND tone	Rogers-Satchell	1	−12.691	0.0000	−11.987	1.0000
GARCHAND tone	Rogers-Satchell	5	−5.956	0.0000	−5.540	1.0000
GARCHAND tone	Rogers-Satchell	22	−0.785	0.4323	−0.335	0.6312
GARCHAND log_artg	Parkinson	1	−4.534	0.0000	−4.098	1.0000
GARCHAND log_artg	Parkinson	5	−2.687	0.0072	−2.457	0.9930
GARCHAND log_artg	Parkinson	22	−0.922	0.3565	−0.640	0.7388
GARCHAND log_artg	Rogers-Satchell	1	−4.221	0.0000	−3.776	0.9999
GARCHAND log_artg	Rogers-Satchell	5	−2.521	0.0117	−2.285	0.9888
GARCHAND log_artg	Rogers-Satchell	22	−0.834	0.4045	−0.552	0.7095
GARCHND $\kappa=30\%$ tone	Parkinson	1	−4.436	0.0000	−4.211	1.0000

GARCHND $\kappa=30\%$ tone	Parkinson	5	-3.798	0.0001	-3.723	0.9999
GARCHND $\kappa=30\%$ tone	Parkinson	22	-2.770	0.0056	-2.804	0.9975
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	1	-4.236	0.0000	-3.999	1.0000
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	5	-3.644	0.0003	-3.558	0.9998
GARCHND $\kappa=30\%$ tone	Rogers-Satchell	22	-2.751	0.0059	-2.772	0.9972
GARCHND $\kappa=30\%$ log_artg	Parkinson	1	-3.152	0.0016	-2.351	0.9906
GARCHND $\kappa=30\%$ log_artg	Parkinson	5	-1.996	0.0459	-1.580	0.9430
GARCHND $\kappa=30\%$ log_artg	Parkinson	22	-1.376	0.1690	-1.059	0.8552
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	1	-3.283	0.0010	-2.454	0.9929
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	5	-2.123	0.0338	-1.717	0.9570
GARCHND $\kappa=30\%$ log_artg	Rogers-Satchell	22	-1.422	0.1551	-1.112	0.8670
GARCHND $\kappa=50\%$ tone	Parkinson	1	+4.102	0.0000	+4.792	0.0000
GARCHND $\kappa=50\%$ tone	Parkinson	5	+1.228	0.2193	+1.813	0.0349
GARCHND $\kappa=50\%$ tone	Parkinson	22	-2.336	0.0195	-1.057	0.8547
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	1	+4.118	0.0000	+4.800	0.0000
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	5	+1.150	0.2501	+1.750	0.0401
GARCHND $\kappa=50\%$ tone	Rogers-Satchell	22	-2.370	0.0178	-1.073	0.8583
GARCHND $\kappa=50\%$ log_artg	Parkinson	1	+6.775	0.0000	+7.009	0.0000
GARCHND $\kappa=50\%$ log_artg	Parkinson	5	+3.300	0.0010	+3.342	0.0004
GARCHND $\kappa=50\%$ log_artg	Parkinson	22	+0.996	0.3195	+1.157	0.1237
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	1	+7.088	0.0000	+7.261	0.0000
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	5	+3.300	0.0010	+3.341	0.0004
GARCHND $\kappa=50\%$ log_artg	Rogers-Satchell	22	+0.983	0.3254	+1.144	0.1264
EGARCH-X	Parkinson	1	-1.048	0.2948	+0.967	0.1668
EGARCH-X	Parkinson	5	+0.423	0.6724	+1.312	0.0948
EGARCH-X	Parkinson	22	+1.608	0.1078	+1.839	0.0330
EGARCH-X	Rogers-Satchell	1	-0.657	0.5111	+1.334	0.0912
EGARCH-X	Rogers-Satchell	5	+0.614	0.5392	+1.454	0.0730
EGARCH-X	Rogers-Satchell	22	+1.601	0.1093	+1.827	0.0339

Verdict: a clean separation emerges across the nine augmented candidates. The simple symmetric and asymmetric news models are dominated by the baseline at every horizon and under both RV proxies. GARCH-X (tone, log_artg) and GARCHAND (tone, log_artg) post strongly negative DM and one-sided Clark–West p-values > 0.99 at h = 1 and h = 5: adding tone² or its asymmetric counterpart purely as a variance shifter destroys forecast accuracy, with GARCHAND tone being the worst performer (DM $\approx$ -12 at h = 1). The GARCHND $\kappa = 30\%$ specifications also lose to the baseline — the 30 %-annualised volatility threshold triggers d_3 too often, so the γ -term injects noise rather than information. Only the GARCHND $\kappa = 50\%$ pair adds OOS value, and only in the news regime they were designed for: $\kappa = 50\%$ with log_art growth beats the baseline at h = 1 and h = 5 with high significance under both proxies (DM $\approx$ +6.8–7.1 at h = 1; +3.3 at h = 5; CW p < 0.001 at both horizons), and is at worst neutral at h = 22 (CW p $\approx$ 0.12). $\kappa = 50\%$ with tone beats the baseline decisively at h = 1 (DM $\approx$ +4.1, CW p < 0.001), is borderline at h = 5 (CW p $\approx$ 0.04 but DM not significant), and turns negative at h = 22. EGARCH-X is roughly tied with the baseline at h = 1 and h = 5 (insignificant DM, weakly positive CW) and posts a marginal positive CW edge at h = 22 (p $\approx$ 0.03), suggesting log-variance plus a small leverage term captures something the

squared-variance specifications miss at the monthly horizon. Taken together, the picture is consistent with a short-lived news shock that is informative only when the previous day was already in a high-volatility regime — exactly the d_3 switch the $\kappa = 50\%$ GARCHND was designed around — and is otherwise noise.

Check 2 — Placebo / regime-only test

Two competing specifications are estimated for each GARCHND $\kappa=50\%$ model: (a) the regime-only model $\gamma \cdot d_3$ (no x^2 term), and (b) five placebos replacing the real x_{t-1} with independent random permutations of it. If the original model does not beat these, the γ estimate reflects regime mean-reversion, not news.

x = tone

Specification	LL	AIC	BIC	γ	SE(γ)	t(γ)
original	-5950.47	11910.94	11940.56	-1.0966	0.1445	-7.59
regime-only ($\gamma \cdot d_3$)	-5950.18	11910.37	11939.99	-1.1218	0.0172	-65.10
placebo seed=2026	-5952.05	11914.09	11943.72	-0.6727	0.0689	-9.76
placebo seed=2027	-5953.47	11916.95	11946.57	-0.6057	0.0498	-12.17
placebo seed=2028	-5950.12	11910.23	11939.86	-0.7363	0.0256	-28.73
placebo seed=2029	-5954.64	11919.28	11948.90	-0.4831	0.1141	-4.23
placebo seed=2030	-5951.64	11913.29	11942.91	-0.8069	0.2155	-3.75

Verdict: original does NOT beat the regime-only / placebo baselines (ΔLL vs regime-only = -0.29 ; placebo LL max = -5950.12 vs original -5950.47). News effect is likely a regime artefact.

x = log_art_growth

Specification	LL	AIC	BIC	γ	SE(γ)	t(γ)
original	-5954.03	11918.06	11947.68	-0.9972	0.1635	-6.10
regime-only ($\gamma \cdot d_3$)	-5950.18	11910.37	11939.99	-1.1218	0.0172	-65.10
placebo seed=2026	-5956.31	11922.63	11952.25	-0.0293	0.3521	-0.08
placebo seed=2027	-5954.29	11918.58	11948.21	-0.5808	0.1936	-3.00
placebo seed=2028	-5955.55	11921.10	11950.73	-0.3951	0.6144	-0.64
placebo seed=2029	-5949.77	11909.54	11939.17	-1.6377	0.1975	-8.29
placebo seed=2030	-5956.19	11922.38	11952.01	-0.2157	0.1938	-1.11

Verdict: original does NOT beat the regime-only / placebo baselines (ΔLL vs regime-only = -3.85 ; placebo LL max = -5949.77 vs original -5954.03). News effect is likely a regime artefact.

Check 3 — SE stability across sigmoid sharpness K

Both GARCHND $\kappa=50\%$ models are refitted at $K \in \{5, 10, 20, 50\}$ and with a hard binary indicator ($K \rightarrow \infty$). Stability is measured by the coefficient of variation of γ across K values; $CV < 0.2$ is considered stable.

x = tone (CV = 0.342 — moderately sensitive to K)

K	LL	γ	SE(γ)	t(γ)	converged
K=5	-5950.65	-1.0209	0.1238	-8.24	True
K=10	-5950.57	-1.0253	0.1020	-10.05	True
K=20	-5950.47	-1.0966	0.1445	-7.59	True
K=50	-5950.59	-0.9086	0.0248	-36.65	True
hard d3	-5954.05	-0.2857	0.0001	-2318.55	True

x = log_art_growth (CV = 0.295 — moderately sensitive to K)

K	LL	γ	SE(γ)	t(γ)	converged
K=5	-5954.08	-1.1225	0.3712	-3.02	True
K=10	-5954.01	-1.1592	0.1874	-6.19	True
K=20	-5954.03	-0.9972	0.1635	-6.10	True
K=50	-5953.87	-1.2235	0.0888	-13.78	True
hard d3	-5955.11	-0.4223	0.0001	-6100.89	True

Verdict: γ keeps its sign and roughly its magnitude across smooth $K \in \{5, 10, 20, 50\}$ for both specifications (tone: ≈ -0.91 to -1.10 ; log_artg: ≈ -1.00 to -1.22), with coefficients of variation in the 0.30 range — neither stable (< 0.2) nor unstable (> 0.5). The hard-indicator fit, by contrast, collapses to a degenerate Hessian (condition number 10^9 – 10^{10} , $SE \approx 10^{-4}$) and yields the implausible t-ratios in the thousands — a clear sign that the binary d_3 kills identification. The smoothed version ($K = 20$, the default) is the right estimator; the t-ratios it reports are the ones to trust.

Check 4 — High-volatility episode clusters ($\kappa=50\%$)

Days where $\sigma_{t-1}^2 \geq \kappa_{\text{high}}$ (annualised 50% vol) are grouped into episodes separated by gaps of at most 5 days. The longest episode determines the recommended block-bootstrap length and HAC bandwidth.

x = tone: 125 triggered days (4.52%), 17 distinct episodes; recommended block length $\geq$ 37 days

#	Start	End	Days
1	2021-02-10	2021-04-05	37
2	2020-03-11	2020-04-21	29
3	2025-10-15	2025-11-13	22
4	2026-02-03	2026-02-18	11
5	2018-12-24	2019-01-02	6
6	2022-11-08	2022-11-15	6
7	2025-04-11	2025-04-17	5
8	2026-03-05	2026-03-11	5
9	2026-03-25	2026-03-30	4
10	2021-01-11	2021-01-13	3

x = log_art_growth: 137 triggered days (4.95%), 21 distinct episodes; recommended block length $\geq$ 37 days

#	Start	End	Days
1	2021-02-04	2021-03-29	37
2	2020-03-04	2020-04-23	36
3	2025-10-15	2025-11-12	21
4	2026-03-05	2026-03-31	19
5	2024-09-30	2024-10-14	11
6	2026-02-03	2026-02-18	11
7	2018-12-24	2019-01-08	10
8	2025-04-11	2025-04-22	7
9	2022-11-08	2022-11-14	5
10	2021-01-07	2021-01-12	4

Check 5 — Lead-lag cross-correlation: news vs realised volatility (Notebook 06)

Model-free test of the maintained hypothesis behind the whole thesis: does news actually *lead* realised volatility, or only move with it contemporaneously? We report the cross-correlation $\rho(k) = \text{Corr}(x_{t+k}, \ln RV_t)$ for $k \in [-10, +10]$, where **$k < 0$ means news LEADS RV** (forward-looking information) and $k > 0$ means RV leads news (reactive coverage). Because both series are autocorrelated, the raw correlations are biased toward significance; the valid test uses the **prewhitened** columns (each series filtered by an AR(10) before correlating), with the white-noise band $\pm 1.96/\sqrt{N} \approx \pm 0.037$. Three regressors are screened: signed **tone**, news intensity **|tone|**, and **log_art_growth**.

Cross-correlations $\rho(k)$ ($k < 0 \Rightarrow$ news leads RV; prewhitened columns are the valid ones)

k	tone (raw)	tone (pw)	tone (raw)	tone (pw)	log_artg (raw)	log_artg (pw)
-10	0.025	0.002	0.008	0.013	0.001	0.010
-9	0.016	-0.012	-0.006	-0.003	0.042	0.024
-8	0.045	0.028	-0.025	-0.024	-0.029	-0.009
-7	0.016	-0.019	0.014	0.022	-0.003	0.008
-6	0.058	0.043	-0.007	-0.007	0.003	0.020
-5	0.015	-0.023	0.002	0.002	0.016	0.027
-4	0.051	0.036	0.002	0.006	0.001	0.020
-3	0.024	0.007	0.023	0.032	0.035	0.033
-2	-0.004	-0.029	-0.040	-0.047	-0.022	0.005
-1	-0.007	-0.028	-0.002	0.001	-0.014	-0.009
0	0.009	-0.006	0.011	0.018	-0.014	-0.006
1	-0.006	-0.032	0.001	0.002	0.029	0.016
2	0.023	0.005	0.017	0.018	0.005	0.003
3	0.035	0.008	0.034	0.041	-0.043	-0.012
4	0.062	0.046	-0.004	-0.005	0.031	0.020
5	0.054	0.035	-0.042	-0.045	0.002	0.016
6	0.023	-0.007	-0.034	-0.033	0.017	0.012
7	0.035	0.020	-0.018	-0.012	0.008	0.001
8	0.008	-0.019	0.001	-0.001	-0.018	-0.018
9	0.025	0.009	-0.001	-0.003	-0.013	-0.034
10	0.015	-0.003	0.013	0.020	-0.014	-0.034

Verdict: under the valid prewhitened band (± 0.037), the maintained "news leads volatility" hypothesis survives only for signed tone, and only weakly: its strongest lead is at $k = -6$ ($\rho_{pw} = +0.043$, just above the band), with a single significant lead lag and $\rho_{pw}(-1) = -0.028$ (below the band). This is consistent with a small, slow forward-looking component in directional sentiment. News intensity |tone| shows no clean lead (its only above-band lead, $k = -2$, is negative, and the significant correlations sit on the lag side $k = +3, +5$ — i.e. RV driving coverage, not the reverse). log_art_growth has no significant lead or lag lags at all once prewhitened. Overall the lead-lag evidence for a genuine news→volatility channel is marginal and confined to signed tone — coherent with the in-sample and OOS results, where the only predictive content came from the volatility-regime switch rather than raw news flow.

Check 6 — HAC and Moving-Block-Bootstrap SEs on γ (Notebook 07)

Sandwich (HAC) and moving-block-bootstrap (MBB, $B=2000$) standard errors on γ for both GARCHND $\kappa=50\%$ specifications, at block lengths $L \in \{25, 50, 75, 100\}$ days (informed by the episode clusters in Check 4).

x = tone

SE method	γ	SE(γ)	t(γ)
iid sandwich	-1.0966	0.1445	-7.59
HAC L=25	-1.0966	0.1360	-8.06
HAC L=50	-1.0966	0.1400	-7.83
HAC L=75	-1.0966	0.1413	-7.76
HAC L=100	-1.0966	0.1411	-7.77
MBB L=25	-1.0966	0.1337	-8.20
MBB L=50	-1.0966	0.1416	-7.74
MBB L=75	-1.0966	0.1403	-7.82
MBB L=100	-1.0966	0.1448	-7.57

x = log_art_growth

SE method	γ	SE(γ)	t(γ)
iid sandwich	-0.9972	0.1635	-6.10
HAC L=25	-0.9972	0.1444	-6.91
HAC L=50	-0.9972	0.1421	-7.02
HAC L=75	-0.9972	0.1453	-6.86
HAC L=100	-0.9972	0.1458	-6.84
MBB L=25	-0.9972	0.1496	-6.67
MBB L=50	-0.9972	0.1469	-6.79
MBB L=75	-0.9972	0.1479	-6.74
MBB L=100	-0.9972	0.1489	-6.69

Verdict: HAC and moving-block-bootstrap corrections leave the t-ratios essentially unchanged from the iid sandwich (tone: $t \approx -7.6$ to -8.2 across all block lengths; log_artg: $t \approx -6.1$ to -7.0). If anything, the HAC and MBB SEs are slightly tighter than iid at moderate L, because the news shocks deliver a partly mean-reverting score that under-weights the long-run variance estimate. The in-sample significance of γ in the $\kappa = 50\%$ GARCHND specifications is robust to plausible serial-correlation corrections, consistent with the favourable rolling-window OOS evidence from Check 1.

Check 7 — GDELT coverage of REMX / rare earths (Notebook 07)

The GDELT article cache (314,048 articles, 2015-04-01 to 2026-04-01) was built with a permissive 19-concept filter (any single match passes). This check quantifies how many articles are genuinely related to the REMX economy across four progressively tighter filter tiers.

Articles per concept (full cache)

Concept	N articles	Share
has_cerium	186,346	59.34%
has_rare_earth	92,383	29.42%
has_lynas	26,066	8.30%
has_samarium	5,005	1.59%
has_neodymium	2,960	0.94%
has_mp_materials	2,891	0.92%
has_praseodymium	2,124	0.68%
has_arafura	1,730	0.55%
has_europium	1,153	0.37%
has_terbium	1,143	0.36%
has_gadolinium	1,051	0.33%
has_yttrium	1,030	0.33%
has_lanthanum	877	0.28%
has_china_northern	702	0.22%
has_dysprosium	576	0.18%
has_ndfeb	505	0.16%
has_ndpr	454	0.14%
has_remx	150	0.05%
has_jl_mag	28	0.01%

Filter tiers

Tier	N articles	Art/day	% zero-days	Mean tone	Std tone
Broad (≥ 1 concept, used in model)	314,048	78.5	0.5%	-0.280	3.299
Medium (metal OR company)	231,708	57.9	0.5%	-0.453	3.585
Tight (rare_earth AND metal/co.)	10,158	2.5	38.5%	-0.130	2.133
Ultra-tight (metal AND company)	559	0.14	91.4%	+0.638	1.850